

VIC TUTORIAL CONT.

GLY606 Water Data Analysis & Modeling

Oct 23rd 2024



Directly go to
<https://jupyterhub.cuahsi.org/>

Let's use “SUMMA Modeling” environment

Server Options

○ Python v3.9.7 - JupyterLab Interface

This is a general-purpose scientific computing environment built on Python version 3.9.7. When using this server, you will be working through the JupyterLab interface. It comes with pre-configured, widely used scientific libraries including scipy, pandas, geopandas, sympy, scikit-learn, matplotlib, ulmo, landlab, gdal, xarray, TauDEM, hscient, dataretrieval, rasterio, pyproj, fiona, geemap, fsspec, and others.

○ R v4.2.3 - JupyterLab Interface

This serves as a general-purpose scientific computing environment built on the R programming language. When using this server, you will have access to both the JupyterLab and RStudio interfaces. The environment is pre-configured with commonly used tools and libraries, including randomForest, forecast, sf, rjsonio, ncdf4, ggmap, dataRetrieval, and WaterML.

● SUMMA Modeling

This is a computing environment built on Python version 3.9 and pre-installed tools to support using the Structure for Unifying Multiple Modeling Alternative (SUMMA). SUMMA is a hydrologic modeling framework that can be used to configure a wide range of hydrological model analysis. When using this server, you will be working through the JupyterLab interface. The environment comes with limited pre-installed scientific Python packages. For more information on the SUMMA model, visit the project [webpage](#).

○ WRF-Hydro v5.1 - NWM v2.0

This computing environment is structured around Python version 3.9 and is tailored for seamless operation with the WRF-Hydro model because it has the source code pre-compiled. WRF-Hydro is a community hydrological model suitable for a range of projects, including real-time flash flood prediction, long-term water supply projections, and evaluations of regional hydroclimate impacts. Notably, WRF-Hydro has been used as the core hydrological module within the National Water Model (NWM) versions 2.0 and 2.1. When using this server, you will be working through the JupyterLab interface. While the environment features a curated selection of pre-installed scientific Python packages, users can follow CUAHSI's guideline to create a customized conda environment, and then install specific Python packages that cater to their unique project requirements. For more information on the WRF-Hydro model, visit the project [website](#).

How do we run VIC?

1

Download source code

- A taste of GitHub

2

Compile source code

- Correct path to the NetCDF library

3

Download example dataset

- We can use GitHub to download it as well

4

Prepare configuration file

- Relative path or absolute path?

5

Run the model!

Step 4: Prepare configuration file



Prepare configuration file

- Relative path or absolute path?

Step 4: Prepare configuration file

4.1. Go to “~/model/example” directory, create a directory named “output” to place model output. Then go to “output” directory, create a directory named “log” to store the log file

```
$ cd ~/model/example
$ mkdir output
$ cd output
$ mkdir log
```

\$ _ jovyan@jupyter-yifancheng X

\$ _ jovyan@jupyter-yifancheng X

```
(base) jovyan@jupyter-yifancheng-5fub:~/model/example/config$ cd ~/model/example
(base) jovyan@jupyter-yifancheng-5fub:~/model/example$ mkdir output
(base) jovyan@jupyter-yifancheng-5fub:~/model/example$ cd output
(base) jovyan@jupyter-yifancheng-5fub:~/model/example/output$ mkdir log
(base) jovyan@jupyter-yifancheng-5fub:~/model/example/output$
```

Step 4: Prepare configuration file

4.2. Go to “~/model/example” directory, create a directory named “config” to place configuration files, and go to “config/” directory

```
$ cd ~/model/example
$ mkdir config
$ cd config
```

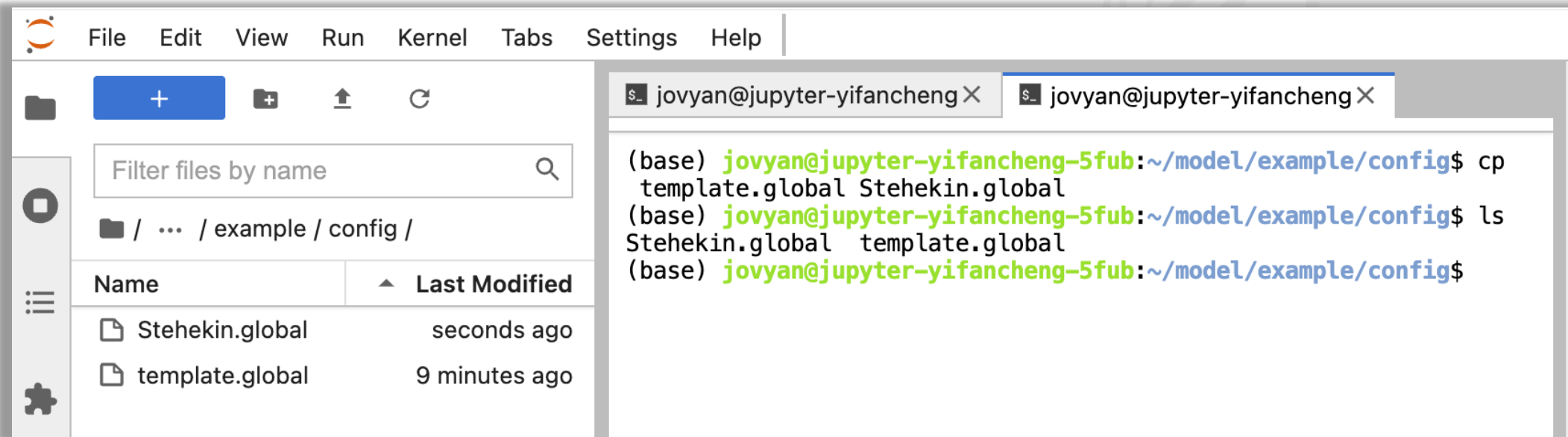
\$_ jovyan@jupyter-yifancheng X

```
(base) jovyan@jupyter-yifancheng-5fub:~$ cd ~
(base) jovyan@jupyter-yifancheng-5fub:~$ cd model/example
(base) jovyan@jupyter-yifancheng-5fub:~/model/example$ ls
data
(base) jovyan@jupyter-yifancheng-5fub:~/model/example$ mkdir config
(base) jovyan@jupyter-yifancheng-5fub:~/model/example$ cd config
(base) jovyan@jupyter-yifancheng-5fub:~/model/example/config$ █
```

Step 4: Prepare configuration file

4.3. Go to github page (GLY606/in_class_practice/model/template.global), download the template file to “config” folder, and make a copy of it and rename it as “Stehekin.global”

```
$ cp template.global Stehekin.global
```



The screenshot displays the JupyterLab interface. On the left, the file explorer shows the directory structure: `/ ... / example / config /`. It lists two files: `Stehekin.global` (modified seconds ago) and `template.global` (modified 9 minutes ago). On the right, the terminal window shows the execution of the command `cp template.global Stehekin.global` and the subsequent `ls` command, which confirms the presence of both files in the directory.

```
(base) jovyan@jupyter-yifancheng-5fub:~/model/example/config$ cp
template.global Stehekin.global
(base) jovyan@jupyter-yifancheng-5fub:~/model/example/config$ ls
Stehekin.global  template.global
(base) jovyan@jupyter-yifancheng-5fub:~/model/example/config$
```

Step 4: Prepare configuration file

4.4. Let's first take a look at the configuration file!

```
$ vim Stehekin.global
```

Necessary for all runs

- Simulation parameters
- Domain file
- Parameter file
- Meteorological forcing files
- Output directory

Required for reasonable configuration

- Write restart file
- Initial condition file

Advanced settings

Step 4: Prepare configuration file

4.4. Let's first take a look at the configuration file!

```
$ vim Stehekin.global
```

1. Simulation parameters

2. Parameter files

3. Domain file

4. Meteorological forcing file

5. Output directory

Relative path v.s. Absolute path

Why does it matter?

- It decides whether our “**configuration file**” can **direct** the code to find the **correct files**!

Example: What is <INSERT_YOUR_FORCING_DIR>?

```
#####
# Forcing Files and Parameters
# netcdf forcing files will be of the form: <FORCING1>YYYY.nc
#####
FORCING1      <INSERT_YOUR_FORCING_DIR>/Stehekin_image_test.forcings_10days.
FORCE_TYPE    AIR_TEMP    tas      # Average air temperature, K
FORCE_TYPE    PREC        prcp     # Total precipitation (rain and snow), kg/m2/s
FORCE_TYPE    PRESSURE    pres     # Atmospheric pressure, Pa
FORCE_TYPE    SWDOWN      dswrf    # Incoming shortwave, W/m2
FORCE_TYPE    LWDOWN      dlwrf    # Incoming longwave radiation, W/m2
FORCE_TYPE    VP          vp       # Vapor pressure, kPa
FORCE_TYPE    WIND        wind     # Wind speed, m/s
# WIND_H      10.0        # height of wind speed measurement. NOTE: in image driver, this globa
l parameter is only used for precipitation correction (if enabled); wind measurement height over bare soi
l is read from the parameter netCDF file.
```

```
(base) jovyan@jupyter-yifancheng-5fub:~/model/example/config$ pwd
/home/jovyan/data/model/example/config
```

```
(base) jovyan@jupyter-yifancheng-5fub:~/model/example/data/VIC_sample_data/image/Stehekin/parameters$ pwd
/home/jovyan/data/model/example/data/VIC_sample_data/image/Stehekin/parameters
```

Absolute path: `/home/jovyan/data/model/example/data/VIC_sample_data/image/Stehekin/parameters`

How do we find the relative path?

Source directory

```
(base) jovyan@jupyter-yifancheng-5fub:~/model/example/config$ pwd  
/home/jovyan/data/model/example/config
```

Target directory

```
(base) jovyan@jupyter-yifancheng-5fub:~/model/example/data/VIC_sample_data/image/Stehekin/parameters$ pwd  
/home/jovyan/data/model/example/data/VIC_sample_data/image/Stehekin/parameters
```

- Step 1: Between the two paths, find their longest common path
 - `/home/jovyan/data/model/example`
- Step 2: Figure out how many levels of directories that source directory needs to exist.
 - Only one, so we only need to exist once from the source directory!
- Step 3: Add the remaining target directory (full target directory excluding the longest common path path in Step 1)

Relative path:

Exercise

Source directory: `/glade/code/yifanc/work/model/config`

Target directory: `/glade/code/yifanc/work/data/ERA5/historical/precip`

Please write the relative path from source directory to target directory

Relative path: `../../data/ERA5/historical/precip`

Step 4: Prepare configuration file

What can we learn from the previous exercise?

- If you decide to use the relative path, the path is how we direct **from the “configuration file” to the “targeted file”**
- To save hassle, I would recommend using the absolute path!

Practice: Complete the configuration file

- Fill in the path to the files!

```
$ vim Stehekin.global
```

```
#####
# Forcing Files and Parameters
# netcdf forcing files will be of the form: <FORCING1>YYYY.nc
#####
FORCING1 <INSERT_YOUR_FORCING_DIR>/Stehekin_image_test.forcings_10days.
FORCE_TYPE ATP_TEMP atemp # Average air temperature, K
FORCE_TYPE PREC prcp # total precipitation (rain and snow), kg/m2/s
FORCE_TYPE PRESSURE pres # Atmospheric pressure, Pa
FORCE_TYPE SWDOWN dswrf # Incoming shortwave, W/m2
FORCE_TYPE LWDOWN dlwrf # Incoming longwave radiation, W/m2
FORCE_TYPE VP vp # Vapor pressure, kPa
FORCE_TYPE WIND wind # Wind speed, m/s
# WIND_H 10.0 # height of wind speed measurement. NOTE: in image driver, this global
# parameter is only used for precipitation correction (if enabled); wind measurement height over bare soil
# is read from the parameter netCDF file.
```

```
#####
# Land Surface Parameters
#####
PARAMETERS <INSERT_YOUR_PARAM_DIR>/Stehekin_test_params_20160327.nc
SNOW_BAND TRUE
BASEFLOW FALSE
JULY_TAVG_SUPPLIED FALSE
ORGANIC_FRACT FALSE
LAI_SRC FROM_VEGPARAM
NODES 3 # number of soil thermal nodes
```

```
#####
# DOMAIN INFO
#####
DOMAIN <INSERT_YOUR_DOMAIN_FILE>/domain.stehekin.20151028.nc
DOMAIN_TYPE LAT lat
DOMAIN_TYPE LON lon
DOMAIN_TYPE MASK mask
DOMAIN_TYPE AREA area
DOMAIN_TYPE FRAC frac
DOMAIN_TYPE YDIM lat
DOMAIN_TYPE XDIM lon
```

```
#####
# Output Files and Parameters
#####
LOG_DIR <INSERT_YOUR_LOG_DIR> # Log directory path
RESULT_DIR <INSERT_YOUR_OUTPUT_DIR> # Results directory path
```

Practice: Complete the configuration file

- Fill in the path to the files!

```
$ vim Stehekin.global
```

```
#####
# Forcing Files and Parameters
# netcdf forcing files will be of the form: <FORCING1>YYYY.nc
#####
FORCING1 <INSERT_YOUR_FORCING_DIR>/Stehekin_image_test.forcings_10days.
FORCE_TYPE ATP_TEMP tnc # Average air temperature, K
FORCE_TYPE PRFC nrcn # Total precipitation (rain and snow). kg/m2/s
```

```
/home/jovyan/data/model/example/data/VIC_sample_data/image/Stehekin/forcings/Stehekin_image_test.forcings_10days.
```

```
#####
# Land Surface Parameters
#####
PARAMETERS <INSERT_YOUR_PARAM_DIR>/Stehekin_test_params_20160327.nc
SNOW_BAND TRUE
BASEFLOW
```

```
/home/jovyan/data/model/example/data/VIC_sample_data/image/Stehekin/parameters/Stehekin_test_params_20160327.nc
```

```
#####
# DOMAIN INFO
#####
DOMAIN <INSERT_YOUR_DOMAIN_FILE>/domain.stehekin.20151028.nc
DOMAIN_TYPE LAT LAT
```

```
/home/jovyan/data/model/example/data/VIC_sample_data/image/Stehekin/parameters/domain.stehekin.20151028.nc
```

```
#####
# Output Files and Parameters
#####
LOG_DIR <INSERT_YOUR_LOG_DIR> # Log directory path
RESULT_DIR <INSERT_YOUR_OUTPUT_DIR> # Results directory path
```

```
/home/jovyan/data/model/example/output/log/
/home/jovyan/data/model/example/output/
```

IT'S
TIME TO
RUN!!!



Step 5: Run the model

5.1. How to run the model!

```
$ cd ~/model/VIC/vic/drivers/image/  
$ ./vic_image.exe -g ~/model/example/config/Stehekin.global
```

 jovyan@jupyter-yifancheng X  jovyan@jupyter-yifancheng X

```
(base) jovyan@jupyter-yifancheng-5fub:~$ cd ~/model/VIC/vic/drivers/image/  
(base) jovyan@jupyter-yifancheng-5fub:~/model/VIC/vic/drivers/image$ ./vic_image.exe -g ~/model/example/c  
onfig/Stehekin.global
```

Sometimes, we need to split the runs into multiple smaller runs...

Necessary for all runs

- Simulation parameters
- Domain file
- Parameter file
- Meteorological forcing files
- Output directory

Required for reasonable configuration

- Write restart file
- Initial condition file

Advanced settings

Restart file and Initial condition file usually mean the same thing in the context of hydrologic modeling!

This file contains the system's status, such as soil moisture, snowpack conditions, etc.

Initial condition file

Once we have an initial condition file, we can **uncomment** the following line and **specify the file path** to the initial condition file

```
#####  
# State Files and Parameters  
#####  
#INIT_STATE # Initial state path/file  
#STATENAME output/image/Stenekin/states # output state file path/prefix. The time (STATEYEAR,STATEMONTH,  
#STATEDAY,STATESEC) will be appended to the prefix automatically in the format yyyyymmdd.  
#STATEYEAR 1949 # year to save model state  
#STATEMONTH 1 # month to save model state  
#STATEDAY 10 # day to save model state  
#STATESEC 82800 # second to save model state  
#STATE_FORMAT NETCDF4_CLASSIC # State file format, valid options:  
#NETCDF3_CLASSIC, NETCDF3_64BIT_OFFSET, NETCDF4_CLASSIC, NETCDF4
```

How do we take a snapshot of the model state for one specific time?

Here we can specify the STATE-related parameters!

```
#####  
# State Files and Parameters  
#####  
#INIT STATE # Initial state path/file  
#STATENAME output/image/Stehekin/states # Output state file path/prefix. The time (STATEYEAR,STATEMONTH,  
STATEDAY,STATESEC) will be appended to the prefix automatically in the format yyyymmdd.  
#STATEYEAR 1949 # year to save model state  
#STATEMONTH 1 # month to save model state  
#STATEDAY 10 # day to save model state  
#STATESEC 82800 # second to save model state  
#STATE_FORMAT NETCDF4_CLASSIC # State file format, valid options:  
#NETCDF3_CLASSIC, NETCDF3_64BIT_OFFSET, NETCDF4_CLASSIC, NETCDF4
```

How do we take a snapshot of the model state for one specific time?

Let's try to edit the following section in the **Stehekin.global** file and rerun the model!

```
$ vim ~/model/example/config/Stehekin.global
$ ./vic_image.exe -g ~/model/example/config/Stehekin.global
```

```
#####
# State Files and Parameters
#####
INIT_STATE # Initial state path/file
STATENAME /home/jovyan/data/model/example/output/Stehekin.restart_file # Output state file path/prefix. T
he time (STATEYEAR,STATEMONTH,STATEDAY,STATESEC) will be appended to the prefix automatically in the format y
yyymmdd.
STATEYEAR 1949 # year to save model state
STATEMONTH 1 # month to save model state
STATEDAY 10 # day to save model state
STATESEC 82800 # second to save model state
STATE_FORMAT NETCDF4_CLASSIC # State file format, valid options:
#NETCDF3_CLASSIC, NETCDF3_64BIT_OFFSET, NETCDF4_CLASSIC, NETCDF4
```

More advanced settings

Necessary for all runs

- Simulation parameters
- Domain file
- Parameter file
- Output directory

Required for reasonable configuration

- Write restart file
- Initial condition file

Advanced settings

Advanced settings

```
#####  
# Miscellaneous Simulation Parameters  
# Generally these default values do not need to be overridden  
#####
```

```
#####  
# Meteorological Forcing Disaggregation Parameters  
# Generally these default values do not need to be overridden  
#####
```

```
#####  
# Precip (Rain and Snow) Parameters  
# Generally these default values do not need to be overridden  
#####
```

```
#####  
# Turbulent Flux Parameters  
# Generally these default values do not need to be overridden  
#####
```

```
#####  
# Carbon Cycle Parameters  
#####
```

```
#####  
# Soil Temperature Parameters
```

```
#####  
# Energy Balance Parameters  
#####
```

Let's play around with the input and output!!

- Please use **Xarray** to examine the following files
 - Meteorological forcings
`(/home/jovyan/data/model/example/data/VIC_sample_data/image/Stehekin/forcings/Stehekin_image_test.forcings_10days.1949.nc)`
 - Domain file
`(/home/jovyan/data/model/example/data/VIC_sample_data/image/Stehekin/parameters/domain.stehekin.20151028.nc)`
 - Parameter files
`(/home/jovyan/data/model/example/data/VIC_sample_data/image/Stehekin/parameters/Stehekin_test_params_20160327.nc)`